Henry Hsu

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SUMMARY

Backend and infrastructure engineer who scales data pipelines and the test infrastructure that keeps them correct. Scaled an async indexing pipeline 3M to 20M+ chunks (3.5x faster parsing), optimized 140+ APIs (27.7% faster), and cut hotfixes 90%. At Google, built 3 integration test infrastructures (Java, Python) from zero across 400+ suites. M.C.S. at Rice, Dec. 2026.

WORK EXPERIENCE

Software Engineering Intern — Jun. 2026 – Aug. 2026

Google — *Taipei, Taiwan*

- **Test Infrastructure:** Built **3** integration test infrastructures from a **zero** baseline (**Java**, **Python**), onboarding **400+** test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Automated Failure Triage:** Built an LLM-backed service that triages scheduled integration test runs and posts root-cause summaries directly onto issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Reproduction Tooling:** Extended the service to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

Backend Engineer — Dec. 2024 – Aug. 2025

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **API Optimization:** Optimized **140+** RESTful APIs through SQL query refactoring, connection pooling (Elasticsearch, Cohere, OpenAI), and Django caching; cut response time of 13 high-traffic APIs by **27.7%** overall and eliminated N+1 queries
- **CI/CD & Testing:** Built a **GitHub Actions** CI pipeline with pytest unit and E2E coverage, reaching **67%** code coverage from a **zero** baseline; reduced production hotfixes by **90%** initially and **50%** long term; managed database migrations in the CD pipeline
- **Open Source:** Contributed **14** merged pull requests to **LlamaIndex** (**15K+ stars**), fixing bugs and adding features across the AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, and MCP Client integrations

Backend Engineer Intern — Sep. 2024 – Dec. 2024

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* — *Taipei, Taiwan*

- **Data Pipeline & Scaling:** Architected an async processing pipeline with **asyncio**, **Celery**, and batch/generator patterns for multi-level knowledge base indexing, achieving **3.5x faster** parsing while scaling from **3M to 20M+** text chunks

EDUCATION

Rice University - Top 20 U.S. University — Aug. 2025 – Dec. 2026 (Expected)

M.C.S., Computer Science (GPA: 4.00/4.00) — *Houston, TX*

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University — Sep. 2020 – Jun. 2024

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) — *Hsinchu, Taiwan*

Minor: Computer Science (Domain GPA: 4.13/4.30)

PROJECTS

Relational Database Engine | *C++, B+ Tree, Query Optimization, TPC-H* — **Aug. 2025 – Dec. 2025**

- **Storage Engine:** Built a C++ storage layer: an **LRU** buffer manager with pinning, paged records, and a **B+ tree** index with range scans
- **Query Processing:** Implemented hash aggregation, a **sort-merge join** over external merge sort, and a cost-based optimizer that compiles SQL into join plans by cardinality, under which the engine completed **TPC-H** at scale factor 1

Chat Bar - Real-Time Multiplayer Map Game | *C++ (TCP Sockets), JSON, MySQL* — **Dec. 2023 – Jan. 2024**

- **Real-Time Client & Protocol:** Built the **SFML** game client as a 60 FPS loop that drains a non-blocking **TCP** socket every frame, and designed the newline-delimited **JSON** packet protocol it shares with the server
- **Chat Rooms:** Extended two-person chat into multi-person rooms on client and server, broadcasting each message to its room only

AWARDS

Presidential Hackathon Winner (2024, Top 5 Nationally) - Urban noise agent: structured-output LLM pipeline and tool calling

23rd Golden Peak Award (2025) - Outstanding Commercial Product, MaiAgent AI Platform

SKILLS

Languages: Java, C/C++, Python, TypeScript | **Systems:** Distributed Storage, B+ Tree Indexing, Query Optimization, Multi-threading
Backend: Django, FastAPI, Express.js | **APIs:** RESTful, WebSocket, Redis Pub/Sub | **Async:** asyncio, Celery | **Testing:** pytest, JUnit
Databases: PostgreSQL, MySQL, Elasticsearch, Milvus | **DevOps:** GitHub Actions, Docker, git, Nginx | **AWS:** EC2, S3, RDS